

On the Transport of Molecular Information in Sub-Diffusion Media with Reflecting Boundary

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ABSTRACT

This work exploits a one-dimensional (1-D) molecular communication (MC) between two nodes considering the sub-diffusion phenomenon for the propagation of information-carrying molecules (ICMs). We also assume that the transmitter (Tx) is surrounded on both sides by an absorbing receiver (Rx) and a reflecting boundary. In this context, we emanate the expressions for the probability density function (PDF) for the spatial-temporal distribution of molecules in terms of the Mittag-Leffler function (MLF). Further, we also find closed-form expressions for ICMs' survival probability (SP) and first arrival probability (FAP) for the considered setup. Moreover, we also derive a closed-form expression of the first arrival time density (FATD) function, an essential mathematical tool to analyze any MC system. We also investigate the effects of reflecting boundary presence on the MC setup based on bit-error-rate performance. Above all, particle-based and Monte-Carlo simulations are performed to validate our analytical expressions.

CCS CONCEPTS

• **Networks** → **Network simulations**; **Network measurement**; **Network performance analysis**.

KEYWORDS

Anomalous diffusion, molecular communication, sub-diffusion, and bit-error rate

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1 INTRODUCTION

There are varieties of technologies already proposed by various researchers that can be used for communication at the nano/micro-scale, such as acoustic communication [1], electromagnetic communication at tera-Hertz (THz) [2] and molecular communication (MC) [3]. Considering typical biological and health-related applications, e.g., disease diagnosis (DD) and particulate drug delivery (PDD), acoustic and THz communication systems are not the perfect choices from the implementation perspectives. Among all of the paradigms mentioned above, an MC system is the slowest process; nevertheless, it is an *obvious* choice to implement DD and PDD systems in reality.

In an MC system, characteristics of specific molecules known as information-carrying molecules (ICMs) are used to carry the information from a transmitter (Tx) to the receiver (Rx) [4]. The transport of ICMs in an MC channel from the Tx to Rx is characterized by diffusion and drift phenomena. Most of the existing works in the literature have considered the MC channel as a diffusion-based MC channel, where transport of ICMs is solely due to the concentration gradient between two points [5]. Further, the diffusion-based MC channels can be classified as classical-diffusive and anomalous-diffusive channels [6]. In a classical-diffusive channel, the mean square displacement (MSD) of the ICM varies linearly with time t . Whereas diffusion, for which the mean square displacement deviates from this linear time dependence, is called anomalous diffusion. Anomalous diffusion is characterized by an MSD proportional to a power-law dependence, i.e., $MSD \propto t^\gamma$, where γ denotes an anomalous diffusion exponent. In the existing literature of diffusion mathematics, the anomalous diffusion have been studied as sub-diffusion, when $0 < \gamma < 1$ and super-diffusion, when $1 < \gamma < 2$ ($\gamma = 1$ signifies the classical Brownian diffusion); see Fig. 1, and details are provided in [6]. The anomalous diffusion of ICMs in the MC system occurs in several cases, such as the transport of ICMs in spatially disordered systems like plasma and turbulent fluids. Moreover, anomalous diffusion can also be found when MC takes place in biological media with traps, receptor binding sites, or macromolecular crowding [7]. Further, it has also been identified that the transport phenomenon in a magnetic field induced microfluidic channel also governs through anomalous diffusion [8].

Thus, the anomalous diffusion phenomenon is worth studying in the context of an MC system, and numerous works [6, 8–16] have done that considering various issues. For example, the first arrival time density (FATD) for the anomalous-diffusive medium has been obtained in [6] using the Caputo fractional derivative, and the channel has been analyzed in terms of bit error rate. In [9], the authors considered the super-diffusive channel and provided

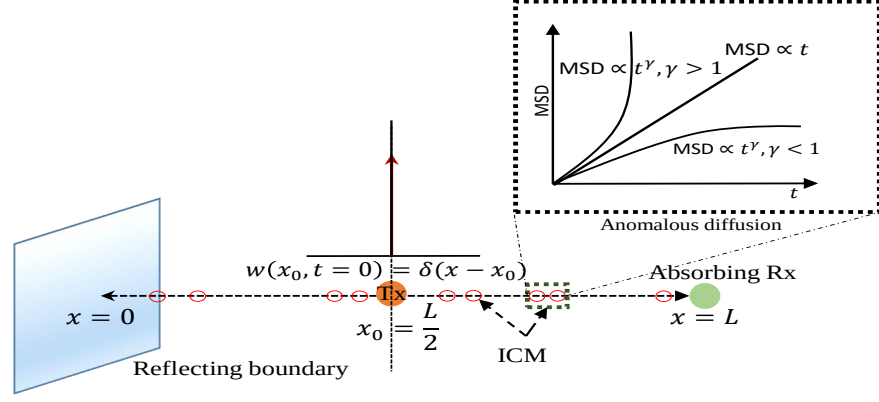


Figure 1: A pictorial presentation of considered system model.

a technique for optimum detection at the Rx. An online event detection method which can cope with sub-diffusion was proposed in [10]. The work [11] has analyzed the stochastic behaviour of an anomalous diffusive channel. However, in [12], the H-diffusion model was used to analyze the anomalous diffusion phenomenon in MC using a timing modulation method. An experimental setup has been provided in [8] for the magnetic field induced micro-channels. The work [13] has introduced a connectivity model for the randomly distributed transmitter and receiver in space. In [15], wherein a one-dimensional (1-D) anomalous-diffusive channel was analyzed considering the scaled-Brownian motion of ICM. Further, a concept of time-dependent drift and diffusion was first introduced in [16] for a 1-D MC channel.

In a few applications, such as MC in biological 1-D media with traps, there are possibilities of the presence of other reflecting surfaces on the intended MC channel apart from the absorbing Rx. Further, this case is also pervasive for the MC system in a magnetic field induced microfluidic channel, and propagation can also be in the semi-infinite domain. Furthermore, note that the arrival probability of ICMs can be increased by including a reflecting wall, resulting in an improvement in the performance of the MC system. Thus, a 1-D MC system having anomalous diffusion and reflecting wall/surface is worth investigating considering the above mentioned practical issues in the various applications.

Therefore based on the above motivations, in this paper, we consider a 1-D semi-infinite anomalous diffusive channel for the study incorporating a perfectly reflecting wall. The expressions for the probability density function (PDF) for spatial-temporal distribution of molecules is derived in terms of MLF. Further, the expressions for survival probability and first absorption/arrival probability of ICMs are derived for the considered model. Moreover, we derive the expression of the FATD function corresponding to the survival probability. We also analyze the MC setup in terms of BER and capacity. In the end, particle-based and Monte-Carlo simulations are performed to analyze the system performance.

2 MC SYSTEM MODEL

We consider an MC setup as shown in Fig. 1, where a point Tx and a point Rx are placed along the x -axis at the positions $x = \frac{L}{2}$ and $x = L$, respectively. A reflecting wall is also placed along the x -axis at the origin $(0, 0, 0)$. Let $w(x, t)$ be the PDF for the molecules is at position x in time t . Further, we assume that the Tx can emit a bunch of molecules as an impulsive transmitting signal, which can be defined as $w(x_0, t = 0) = \delta(x - x_0)$. The ICMs can move in both directions, i.e., towards the Rx or the reflecting wall. We assume that the ICM does not alter their characteristics after colliding with the reflecting wall and returns to its previous position before a collision with the reflecting wall. Rx is assumed to be a perfectly absorbing type which means the ICMs are absorbed once they reach the point $x = L$. For the propagation of ICMs in the channel, we assume only sub-diffusion without drift, in which ICMs can move from high concentration regime to low concentration regime. We also consider that two ICMs never interact, and their propagation is independent of each other.

2.1 Anomalous Diffusion Model

Anomalous diffusion of a molecule is usually characterized by its mean squared displacement, and it is given as [17, Eq. (1)]

$$\langle x^2(t) \rangle \sim \frac{2\mathcal{D}_\gamma}{\Gamma(1+\gamma)} t^\gamma, \quad (1)$$

here $\mathcal{D}_\gamma$ is the generalized diffusion constant associated with the anomalous diffusion exponent γ , and $\Gamma(a)$ is the Gamma function of variable a . An ordinary Brownian diffusion, i.e., when $\mathcal{D}_\gamma \triangleq \mathcal{D}$ and $\gamma = 1$, follows Fick's second law as $x^2(t) \sim t$. When $0 < \gamma < 1$, the process is called sub-diffusive. We consider the sub-diffusive process as a particular case in this paper for analysis. The sub-diffusive processes are omnipresent in nature [18] and are particularly useful for understanding transport phenomena in complex systems such as biological media with traps and also in legend receptor bindings [19, 20]. Note that we use a continuous-time random walk (CTRW) with the waiting-time distribution between the steps that show a long-tail behaviour, i.e., $\Psi(t) \sim \frac{1}{t^{\gamma+1}}$ for $t \rightarrow \infty$ with $0 < \gamma < 1$.

The anomalous diffusion of the molecule can also be characterized by the fractional diffusion equation (FDE), which represents the evolution of the PDF $w(x, t)$ of the molecule at position x and at a time t . The fractional partial differential equation in a 1-D channel is described as [21, Eq. (34)]¹

$$\frac{\partial}{\partial t} w(x, t) = \mathcal{D}_Y {}_0D_t^{1-\gamma} \frac{\partial^2}{\partial x^2} w(x, t), \quad (2)$$

where $\mathcal{D}_Y$ already appears in (1) and ${}_0D_t^{1-\gamma} w(x, t)$ is known as Riemann-Liouville operator which is defined as [23]

$${}_0D_t^{1-\gamma} w(x, t) = \frac{1}{\Gamma(\gamma)} \frac{\partial}{\partial t} \int_0^t \frac{w(x, \tau)}{(t - \tau)^{(1-\gamma)}} d\tau, \quad (3)$$

here τ is a dummy variable.

2.2 Solution of FDE for the Considered Model

In this section, we solve the FDE provided in (2) to get the exact solution for $w(x, t)$ for the considered scenario, as shown in Fig. 1. To do so, first, we define the initial and boundary conditions for the considered system model that are as follows:

- We assume impulsive release from Tx instantly at $t = 0$, i.e., $w(x, t = 0)|_{x=x_0} = \delta(x - x_0)$, where $\delta(\cdot)$ is the Kronecker delta function.
- Rx is perfectly absorbing type that means vanishing concentration at $x = L$, i.e., $w(x, t)|_{x=L} = 0$.
- At the reflecting wall, PDF follows the Neumann boundary condition, i.e., $\frac{\partial}{\partial x} w(x, t)|_{x=0} = 0$. This arises due to the vanishing boundary condition for the derivative of the PDF $w(x, t)$.

Now we use the variables' separation method to get the required solution. Thus, for a given mode n , one can write the final solution of (2) in the form of

$$W_n(x, t) = X_n(x) T_n(t), \quad (4)$$

here $X_n(x)$ and $T_n(t)$ are the spatial and temporal solutions for the n th mode PDF $W_n(x, t)$. Now plugging above solution in (2) we get

$$X_n \frac{\partial T_n(t)}{\partial t} = T_n(t) \mathcal{D}_Y {}_0D_t^{1-\gamma} \frac{\partial^2 X_n(x)}{\partial x^2}. \quad (5)$$

Above equation further can be simplified as

$$\frac{1}{T_n(t) \mathcal{D}_Y} \frac{\partial T_n(t)}{\partial t} = \frac{1}{X_n(x)} {}_0D_t^{1-\gamma} \frac{\partial^2 X_n(x)}{\partial x^2} = l, \quad (6)$$

where l is a constant. Note that the right-hand side (RHS) of above equation is a function of variable x and the left-hand side (LHS) a function of t . Therefore, equality in (6) must be true for the case when both RHS and LHS must be independent of each other, and hence we have introduced a constant l . Further, to find the solution of above set of equations tractable, we chose the constant $l = -(\lambda_{n,Y})^2$, where $\lambda_{n,Y} \in \mathbb{R}$ and hence $l \in \mathbb{R}^-$. Thus, for a given mode of n , merging (4) in (2), the decoupled set of eigen-equations

can be obtained as [21]

$$\frac{1}{\mathcal{D}_Y} \frac{\partial T_n(t)}{\partial t} = -(\lambda_{n,Y})^2 {}_0D_t^{1-\gamma} T_n(t), \quad (7)$$

$$\frac{\partial^2 X_n(x)}{\partial x^2} = -(\lambda_{n,Y})^2 X_n(x), \quad (8)$$

where $\lambda_{n,Y}$ is a fractional eigenvalues. Now, the temporal eigenfunction $T_n(t)$ being governed by the FDE (2) is described by the Mittag-Leffler function² ($E_{\alpha,\beta}(z)$); with complex parameters α and β . Now, the temporal solution $T_n(t)$ is derived as [21]

$$T_n(t) = E_Y(-\mathcal{D}_Y(\lambda_{n,Y})^2 t^\gamma) \triangleq \sum_{j=0}^{\infty} \frac{(-\mathcal{D}_Y(\lambda_{n,Y})^2 t^\gamma)^j}{\Gamma(1 + \gamma j)}, \quad (9)$$

where $E_Y(z)$ is a Mittag-Leffler function with the parameters $\alpha = \gamma$ and $\beta = 1$. Now, the solution for $X_n(t)$ can be obtained using the concept of characteristics equation [25]

$$X_n(x) = C_n \cos(\lambda_{n,Y} x) + d_n \sin(\lambda_{n,Y} x), \quad (10)$$

here C_n and d_n are the arbitrary constants. To get the solution for the above equation incorporating the boundary conditions provided at the beginning of this section, we put $d_n = 0$. Thus, finally $X_n(x)$ can be expressed as

$$X_n(x) = C_n \cos\left(\frac{(2n+1)\pi x}{2L}\right), \quad (11)$$

here $\lambda_{n,Y} = \frac{(2n+1)\pi}{2L}$ and $k \in \{0, 1, 2, \dots\}$. Now, merging (4), (9), and (11), we have the solution for the n th mode as

$$W_n(x, t) = C_n E_Y\left(-\mathcal{D}_Y \left(\frac{(2n+1)\pi}{2L}\right)^2 t^\gamma\right) \cos\left(\frac{(2n+1)\pi x}{2L}\right). \quad (12)$$

The linear combination of $W_n(x, t)$ for various values of n will be the final solution of the FDE (2), and it can be written as

$$w(x, t) = \sum_{n=0}^{\infty} C_n E_Y\left(-\mathcal{D}_Y \left(\frac{(2n+1)\pi}{2L}\right)^2 t^\gamma\right) \cos\left(\frac{(2n+1)\pi x}{2L}\right), \quad (13)$$

here arbitrary constant C_n can be found using the initial condition $w(x = L/2, t = 0) = \delta(x - L/2)$. Now, one can manipulate (13) to get the value of C_n as follows

$$\int_{x=0}^L \delta\left(x - \frac{L}{2}\right) \times \cos\left(\frac{(2n'+1)\pi}{2L} x\right) dx = \sum_{n=0}^{\infty} \int_{x=0}^L C_n \cos\left(\frac{(2n+1)\pi}{2L} x\right) \cos\left(\frac{(2n'+1)\pi}{2L} x\right) dx. \quad (14)$$

Following the orthogonality property of sinusoidal function [25], the integration in RHS side of the above equation is equal to 1, and we get the value of $C_n = \frac{2}{L} \left[\cos\left(\frac{(2n+1)\pi}{2L} \pi\right) \right]$. Finally, the solution

¹It has been observed in various works [22] that the FDE is accurate and valuable for the analysis of biomolecular reactions in sub-diffusive particles.

²The generalized Mittag-Leffler function of argument z is defined as $E_{\alpha,\beta}(z) = \sum_{j=0}^{\infty} \frac{(z)^j}{\Gamma(\beta + \alpha j)}$ [24].

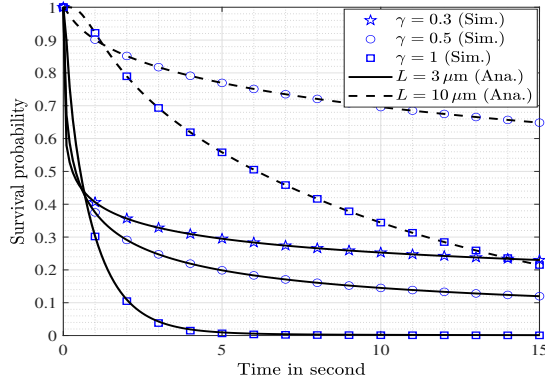


Figure 2: SP as a function of time for different values of γ (Cf. (17)).

becomes as

$$w(x, t) = \frac{2}{L} \sum_{n=0}^{\infty} \cos\left(\frac{(2n+1)\pi}{4}\right) \times \cos\left(\frac{(2n+1)\pi}{2L}x\right) \times E_{\gamma}\left(-\mathcal{D}_{\gamma}\left(\frac{(2n+1)\pi}{2L}\right)^2 t^{\gamma}\right), \quad (15)$$

2.3 First Arrival Probability of ICM

The survival probability $F_{sur}(t)$ is defined as the probability that the sub-diffusive random walker does not reach the Rx by time t . When the Rx is placed at $X = L$, the survival probability of a ICM starting from x_0 , can be obtained as

$$F_{sur}(t) = \int_{x=0}^L w(x, t|x_0) dx. \quad (16)$$

The solution can be obtained as

$$F_{sur}(t) = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{\cos(n\pi)}{(2n+1)} \times \cos\left(\frac{(2n+1)\pi}{4}\right) \times E_{\gamma}\left(-\mathcal{D}_{\gamma}\left(\frac{(2n+1)\pi}{2L}\right)^2 t^{\gamma}\right). \quad (17)$$

Now, the FATD $f_a(t)$ can be defined as the probability that a particle has reached the absorption point L between times t and $t + \Delta t$. It is related with the survival probability as $f_a(t)\Delta t = F_{sur}(t) - F_{sur}(t + \Delta t)$. Using the first order Taylor series approximation, the relation between $F_{sur}(t)$ and $f_a(t)$ can be written as

$$f_a(t) = -\frac{\partial F_{sur}(t)}{\partial t}. \quad (18)$$

Now, applying the identity of differentiation of Mittag-Leffler function which is provided in [26, Eq. 49]

$$f_a(t) = \frac{4\pi\mathcal{D}_{\gamma}}{L^2 t^{1-\gamma}} \sum_{n=0}^{\infty} (2n+1) \cos(n\pi) \times \cos\left(\frac{(2n+1)\pi}{4}\right) \times E_{\gamma,\gamma}\left(-\mathcal{D}_{\gamma}\left(\frac{(2n+1)\pi}{2L}\right)^2 t^{\gamma}\right). \quad (19)$$

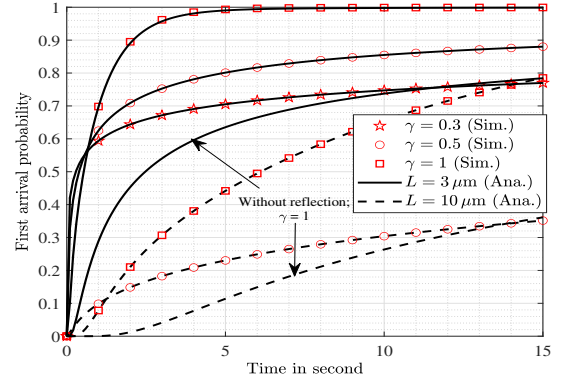


Figure 3: FAP as a function of time for different values of γ (Cf. (20)).

Next, one can obtain the expression for the FAP for the absorbing wall positioned at L m by integrating FATD over the specific time-interval τ as

$$F_a(\tau) = \int_{t=0}^{\tau} f_a(t) dt = \sum_{n=0}^{\infty} \ell_n \left[1 - E_{\gamma}\left(-\mathcal{D}_{\gamma}\frac{(2n+1)^2 \pi^2}{4L^2} \tau^{\gamma}\right) \right], \quad (20)$$

here $\ell_n = \frac{4}{\pi} \frac{\cos(n\pi)}{(2n+1)} \cos\left(\frac{(2n+1)\pi}{4}\right)$.

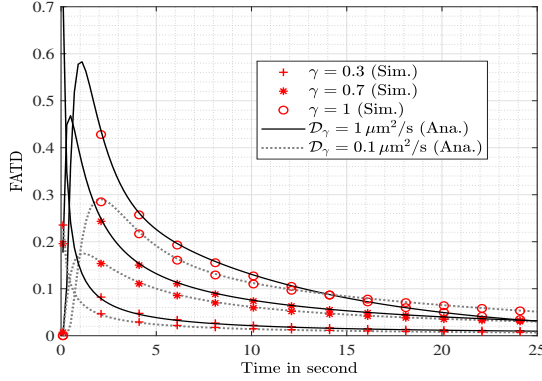
Fig. 2 and 3 present the plots for the SP and FAP of ICM at the Rx, respectively for $\mathcal{D}_{\gamma} = 10 \mu\text{m}^2/\text{s}$. In these figures, we use Monte-Carlo particle simulation to validate the analytical expressions derived in this section; Cf. (17) and (20). The system parameters for these figures are shown in Table 1. We also fix these parameters throughout the paper for analysis in the subsequent chapters until stated otherwise. We use the same methodology as provided in [21, 27] for the simulation. In short, we divide the total time τ in N number of small time-steps $\delta t = \tau/N$. Each of the ICM moves randomly in 1-D channel and shift its position by Δx , which is assumed to be a Normal distributed random variable with $\text{MSD} < x^2(t) > \sim \frac{2\mathcal{D}_{\gamma}}{\Gamma(1+\gamma)} \delta t^{\gamma}$. The new position x_{p+1} is related to its previous position x_p as $x_{p+1} = x_p + \Delta x$. Once an ICM reaches the absorbing Rx, it is counted on the counter and erased from the channel. And if ICM reaches the reflecting wall, it gets back to the position that ICM have before the collision. The above steps continue until the total time duration τ . Using the total counting of ICMs at each time step δt , we can find the arrival probability of ICMs³.

Figs. 2 and 3 present the survival and first arrival probabilities as a function of time for different sets of length (L) of the channel and anomalous diffusion exponent (γ). It can be observed that when the value of γ is high ($\gamma = 0.5$), the arrival probability of ICM is consistently high compared to the point when γ is low ($\gamma = 0.3$). And corresponding SP reduces when we increase the value of γ . Since the high value of γ leads to the faster diffusion of ICMs in the channel. Further, if we make the value of L lower, i.e., for $L = 3 \mu\text{m}$, the value of arrival probability is highest (≈ 1), and the value of

³We use the MATHEMATICA tool to get the analytical values of FAP and SP as the MLF is not available in the latest version of MATLAB. Note that the simulated values perfectly corroborate the figures' analytical values.

Table 1: Table of System Parameters

Parameters	Values
Q_T	$\{200, 500, 1000\}$
$\mathcal{D}_\gamma$	$\in \{0.1, 1, 10\} \mu\text{m}^2/\text{s}^\gamma$
γ	$\in \{0.2, 0.3, 0.5, 0.7, 1\}$
L	$\in \{3, 5, 10, 20\} \mu\text{m}$
x	$0 - L \mu\text{m}$
n	200
τ	(0-15) s

**Figure 4: FATD as a function of time for different values of γ and $\mathcal{D}_\gamma$ for $L = 5 \mu\text{m}$ (Cf. (19)).**

SP is lowest (≈ 0). This is because the presence of a reflecting wall increases the chance of reaching the ICMs at the Rx, eventually reducing the corresponding SP. Interestingly, we find that the values of FAP are higher with a reflecting boundary than the FAP values obtained without a reflecting boundary for any distance between the Tx and Rx.

In Fig. 4, FATD is examined as a function of time for different sets of γ and $\mathcal{D}_\gamma$ using particle simulation. We can observe that the FATD attains its peak at a particular time t , and after that, it starts reducing as the t increases. Herein, the peak value is highest ≈ 0.58 for the set $\mathcal{D}_\gamma = 1 \mu\text{m}^2/\text{s}^\gamma$ and $\gamma = 1$ compared to the all other sets of γ and $\mathcal{D}_\gamma$. This is due to the fact that ICMs diffuse fast for high γ and $\mathcal{D}_\gamma$.

3 TRANSMISSION AND RECEPTION MODEL

We consider a time-slotted communication channel between the Tx and Rx, in which Tx transmits a binary bit in each time slot, and Rx receives the corresponding bit in the same time slot. Let the Tx has binary information sequence $B = \{b[1], b[2], \dots, b[j], \dots, b[K]\}$ of length K , where $b[j] \in \{0, 1\}$, for the transmission to Rx. For the transmission of each bit from Tx, the on-off-keying (OOK) method is adopted, where a few fixed number of molecules (Q_T) are emitted from Tx for the binary bit 1 and Tx remains inactive for bit 0. Subsequently, depending upon the number of molecules received (Q_R) at the Rx, the corresponding bit is decoded at the Rx. Now, let the received information sequence is denoted as $\hat{B} = \{\hat{b}[1], \hat{b}[2], \dots, \hat{b}[j], \dots, \hat{b}[K]\}$. Further, in line with the several

works (e.g., [28, 29] and references therein), there is perfect synchronization between Tx and Rx is assumed, i.e., Tx and Rx have capabilities to identify the start and end of the time slot intervals accurately.

The number of molecules received at the Rx in one time slot duration τ_s depends on their FAP and it can be represented using (20) as

$$P_{j-i} = \sum_{n=0}^{\infty} \ell_n \left[E_\gamma \left(-\mathcal{D}_\gamma \frac{(2n+1)^2 \pi^2}{4L^2} [(j-i)\tau_s]^\gamma \right) - E_\gamma \left(-\mathcal{D}_\gamma \frac{(2n+1)^2 \pi^2}{4L^2} [(j-i+1)\tau_s]^\gamma \right) \right], \quad (21)$$

where P_{j-i} denotes the probability of molecules transmitted in i th bit duration and arrived in j th bit duration. Now, the expected number of molecules at the Rx in j th time-slot is formulated in terms of two hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$ corresponding to the symbols 0 and 1, respectively, as follows

$$\begin{aligned} \mathcal{H}_0 : \Lambda_0[j] &= \sum_{i=1}^{j-1} b[j-i] Q_T P_i + \eta_o q_o, \\ \mathcal{H}_1 : \Lambda_1[j] &= Q_T P_0 + \sum_{i=1}^{j-1} b[j-i] Q_T P_i + \eta_o q_o, \end{aligned}$$

where $b[j] \in \{0, 1\}$, and P_0 and P_i are the first arrival probabilities for the intended and interfering time-slots, respectively. The η_o corresponds to the noise power, which is equivalent to the number of molecules from the unintended source [30] and q_o is the arrival probability of a noise molecule at the Rx. The average BER considering all the bit slots can be formulated as

$$P_e = \frac{1}{K} \sum_{j=1}^K P_e[j]. \quad (22)$$

Hereby, the expression of BER $P_e[j]$ in the j th bit duration can be formulated as

$$P_e[j] = \Pr(b[j] = 0) \Pr(\hat{b}[j] = 1 | b[j] = 0) + \Pr(b[j] = 1) \Pr(\hat{b}[j] = 0 | b[j] = 1), \quad (23)$$

where $\Pr(b[j] = 0) = \beta_0$ and $\Pr(b[j] = 1) = \beta_1$ correspond to the prior probabilities for symbols 0 and 1, respectively, emitted from the Tx. The terms $\Pr(\hat{b}[j] = 1 | b[j] = 0)$ and $\Pr(\hat{b}[j] = 0 | b[j] = 1)$ are expanded as

$$\Pr(\hat{b}[j] = 1 | b[j] = 0) = 1 - \sum_{l=1}^{\lfloor \Theta[j] \rfloor} \frac{\exp(-\Lambda_0[j]) (\Lambda_0[j])^l}{l!}, \quad (24)$$

and,

$$\Pr(\hat{b}[j] = 0 | b[j] = 1) = \sum_{l=1}^{\lfloor \Theta[j] \rfloor} \frac{\exp(-\Lambda_1[j]) (\Lambda_1[j])^l}{l!}, \quad (25)$$

respectively, here $\Theta[j]$ denotes the decision threshold at Rx. To minimize the BER, the optimal value of decision threshold Θ^* can be obtained in such a way that

$$\Theta^*[j] = \min_{\Theta[j]} P_e[j]. \quad (26)$$

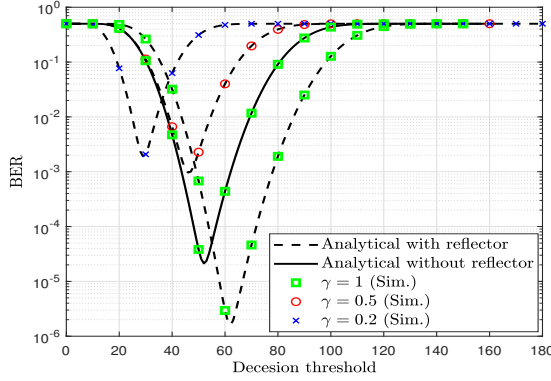


Figure 5: BER as a function of decision threshold for different sets of γ and total channel length for $\tau_s = 5$ s, $L = 3$ μ m and $Q_T = 100$.

Now, to obtain the BER_{min} , the log-likelihood-ratio-test is used as provided in [31] and its expression is given as Next, we can find the value of BER_{min} , by putting the optimal threshold $\Theta^*[j]$ in (23).

4 NUMERICAL RESULTS

For numerical evaluation of the proposed information transmission model, we plotted BER as a function of threshold in Fig. 5 (a) for different sets of L and γ . We use Monte-Carlo simulation for 10^3 iterations to corroborate the analytical results. It is found that for lower γ , BER_{min} is always higher. Moreover, in case of $\gamma = 0.2$, it is very high compare to the $\gamma = 0.5$ and 1. Because, with lower γ , the arrival probability of ICMs becomes low compared to the higher γ . Further, to see the effect of the presence of a reflector in the channel, we compared it with the case when there is no reflecting surface in the medium (general model). It is noted that the existence of a reflector enhances the performance of the MC system, and BER is reduced compared to the case when there is no reflector.

We generate Fig. 6 to observe the simultaneous effects of channel length and the presence of a reflector. In this figure, BER_{min} is plotted as a function of channel length and compared with the case when there is no reflecting surface in the channel (general model). It is observed that BER increases with the increase in the distance for each value of γ , and BER is higher for the case of lower γ . Interestingly, we obtain a lower BER_{min} for a specific range of L in the case of the proposed model compared with the general model.

5 CONCLUSION

This work considered a semi-infinite one-dimensional molecular communication system considering the sub-diffusion phenomenon for transporting information-carrying molecules. A reflecting wall is placed in the channel at the mirrored position of the absorbing receiver, i.e., the transmitter is placed between the receiver node and a reflecting wall. In this context, we derived the expressions for the probability density function for the spatial-temporal distribution of molecules. Further, we found the analytical closed-form expressions of molecules' survival probability and first absorption probability for the considered setup. Moreover, we derived the closed-form

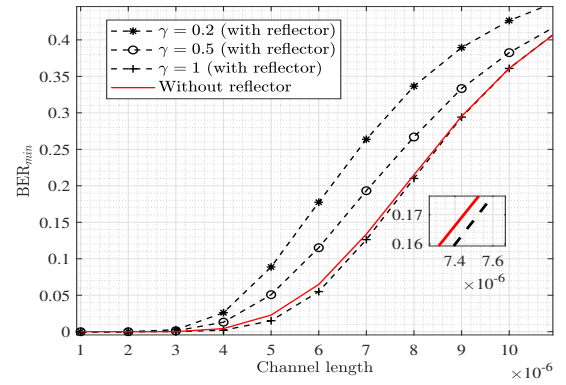


Figure 6: Performance comparison of the channels with and without reflector on the basis of BER_{min} for $Q_T = 100$.

expressions of the first passage time density function. We also analyzed the MC setup in terms of bit-error-rate, and it is investigated that for the case of lower values of anomalous diffusion exponent, BER values are higher compare to case when anomalous diffusion exponent values are higher. Further, we figured out that the presence of a reflector improves the system's performance for a specific range of channel lengths. Our future work involves the optimum positioning of a reflector to enhance an MC system's performance, further considering different channel dimensions.

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